ORIGIN OF LIFE · THEORY

From Rule-Taking *to Rule-Making Chemistry*

A minimal population–environment feedback model

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THE QUESTION

When does chemistry stop adapting to its conditions — *and start authoring them?*

CONTEXT

Classical origin-of-life models

WELL STUDIED

Replicators

Autocatalytic sets
Hypercycles
Protocells

Fidelity, replication, inheritance.

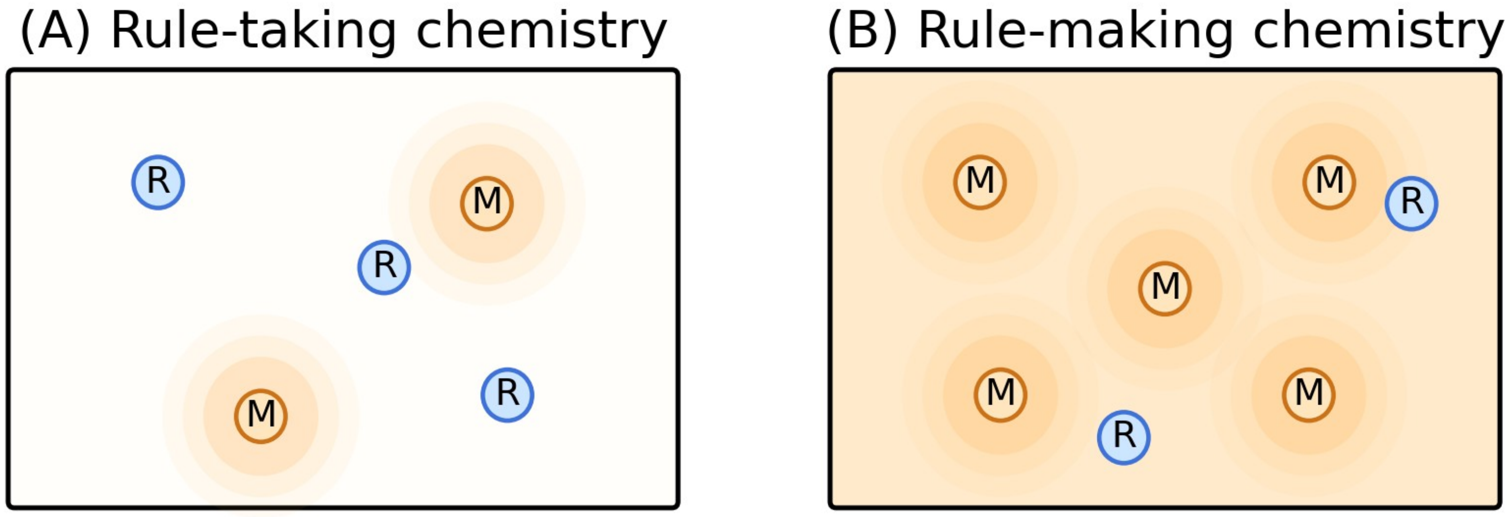
USUALLY FIXED

The environment

Reaction conditions imposed externally
Not shaped by the chemistry it hosts
No feedback into selection

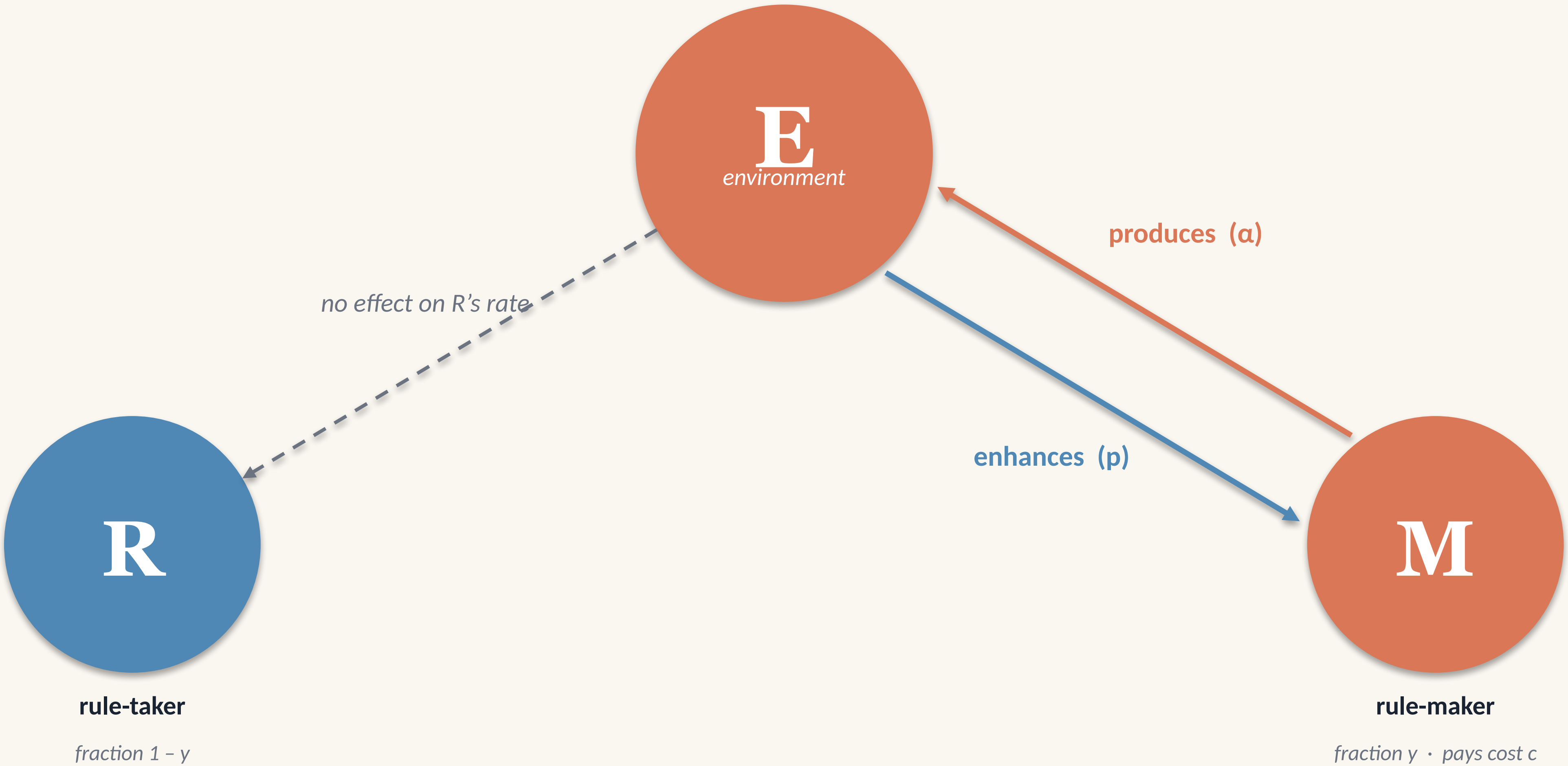
But life is defined by reshaping its surroundings.

Two chemistries



Rule-takers (R) depend on the environment. Rule-makers (M) reshape it — at a cost.

Minimal coupling: population ↔ environment



R sits in E but ignores it ($\pi_R = 1$). Only M's replication rate responds to E.

THE DYNAMICS

Two coupled equations

POPULATION (replicator equation)

$$\dot{y} = y(1-y)\cdot(s-c+pE-1)$$

ENVIRONMENT (production – decay)

$$\dot{E} = \alpha y - \beta E$$

M produces E. A larger E raises M's replication rate — more M, more E.

THE THRESHOLD

One number controls the regime

$$p_{eff} = p\,\alpha / \beta \quad > \quad p_c = c + 1 - s$$

EFFECTIVE FEEDBACK

how strongly modifiers help,
weighted by how long they last

COST BARRIER

what rule-makers give up
to modify the environment

Cross it, and a second stable state appears.

PHASE DIAGRAM

Cost versus feedback strength

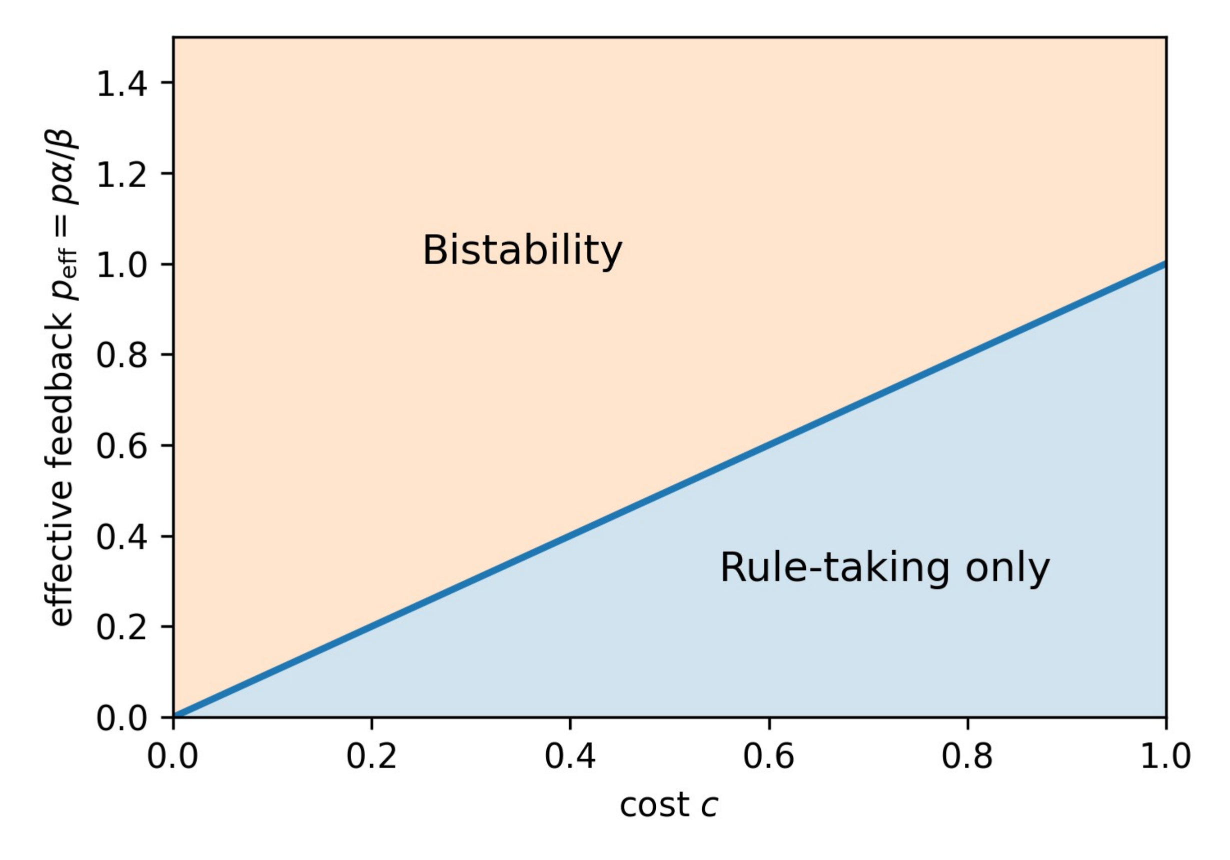
ABOVE THE LINE

Rule-takers and rule-makers each form a stable state.

BELOW THE LINE

Only rule-taking persists — modifiers can't pay for themselves.

Stronger feedback compensates for higher cost — linearly.



GEOMETRY

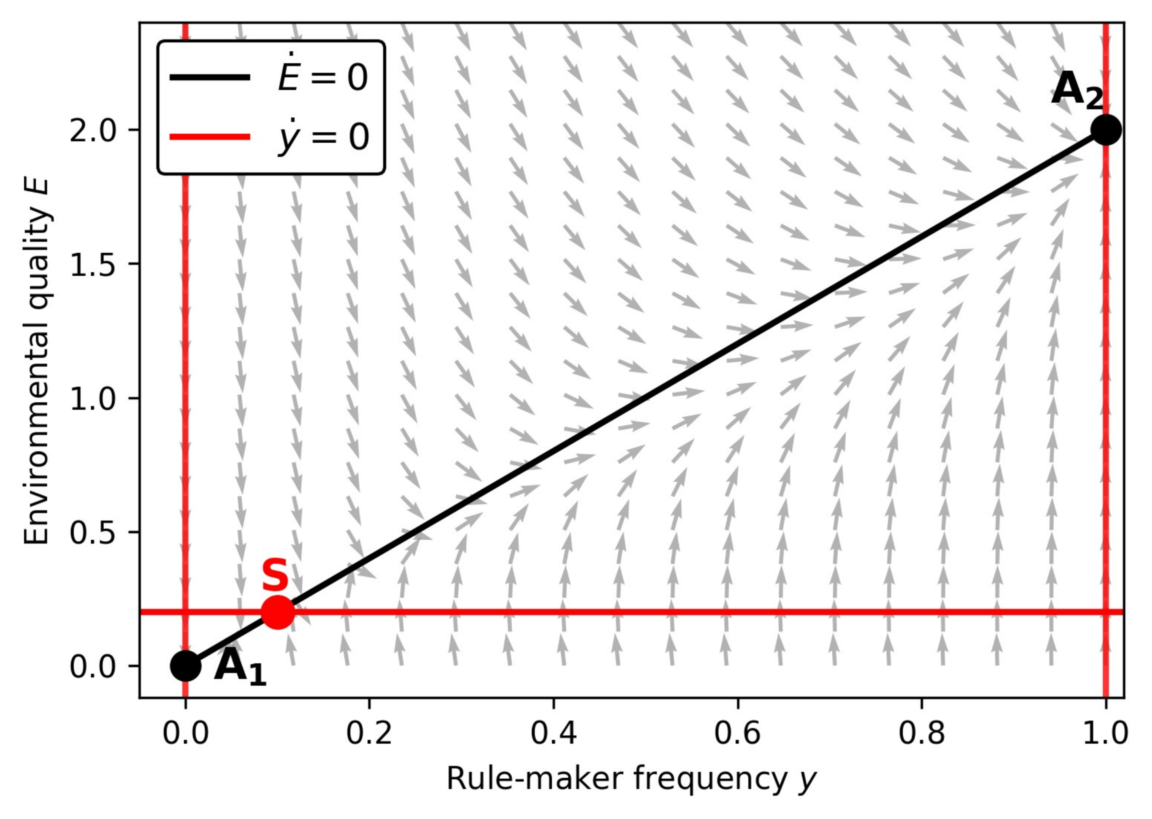
A saddle divides the basin

TWO ATTRACTORS

A_1 = rule-takers only. A_2 = rule-makers dominate.

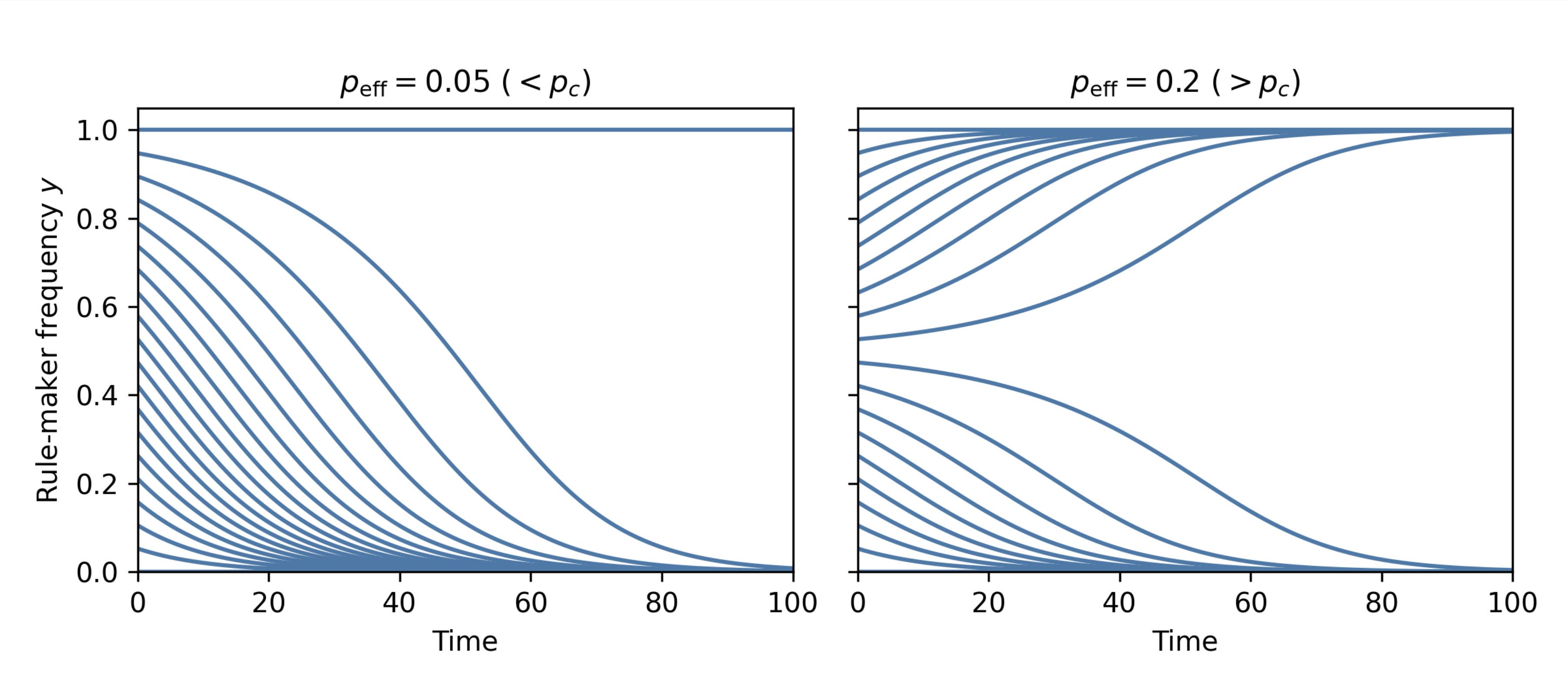
ONE SADDLE

Its stable manifold draws the separatrix between the two fates.



CASE 1 — NO MEMORY

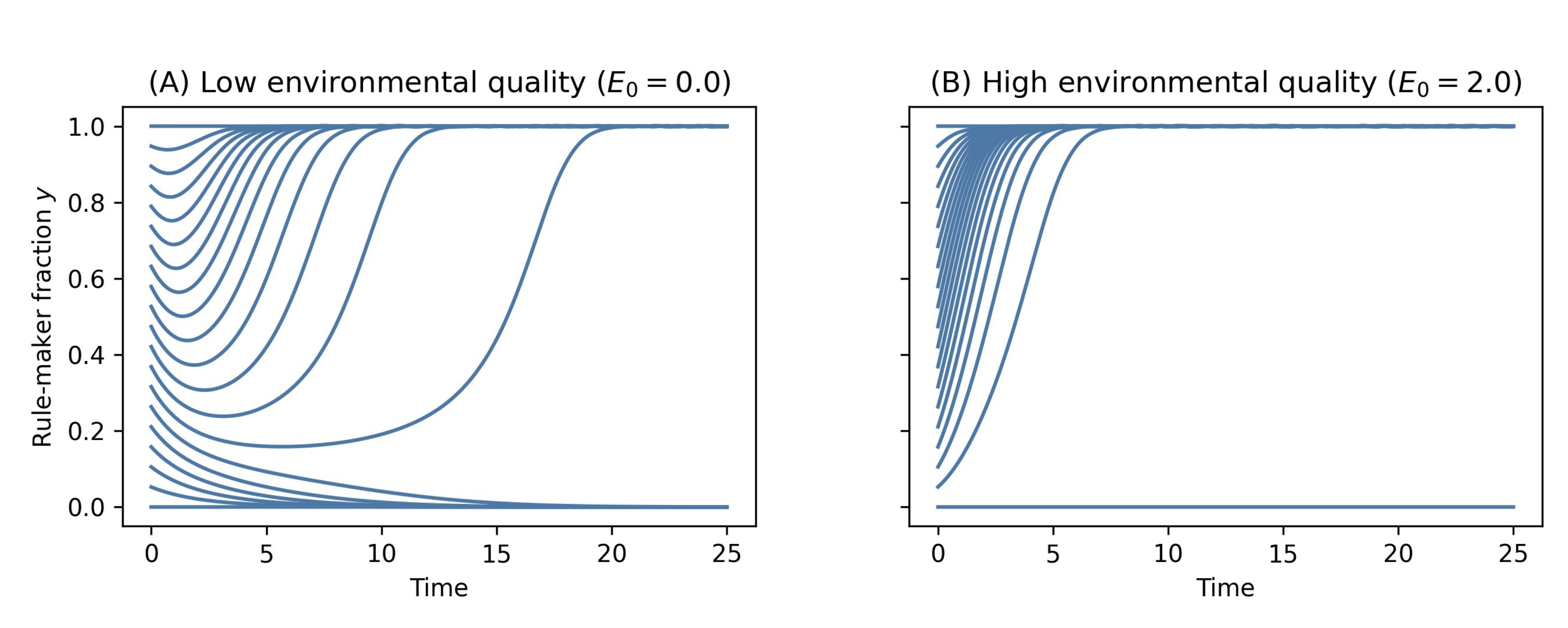
Composition alone decides



When the environment relaxes instantly, only a sharp frequency threshold matters.

CASE 2 — ENVIRONMENTAL MEMORY

History reshapes the basin



Because E remembers, a favourable past lets even rare rule-makers invade.

Same dynamics, real molecules

MINIMAL REACTION NETWORK

$$\mathbf{R} + \mathbf{S} \rightarrow 2\mathbf{R}$$

rule-taker replication

$$\mathbf{M} + \mathbf{S} \rightarrow 2\mathbf{M}$$

rule-maker replication

$$\mathbf{M} + \mathbf{S} + \mathbf{B} \rightarrow 2\mathbf{M} + \mathbf{B}$$

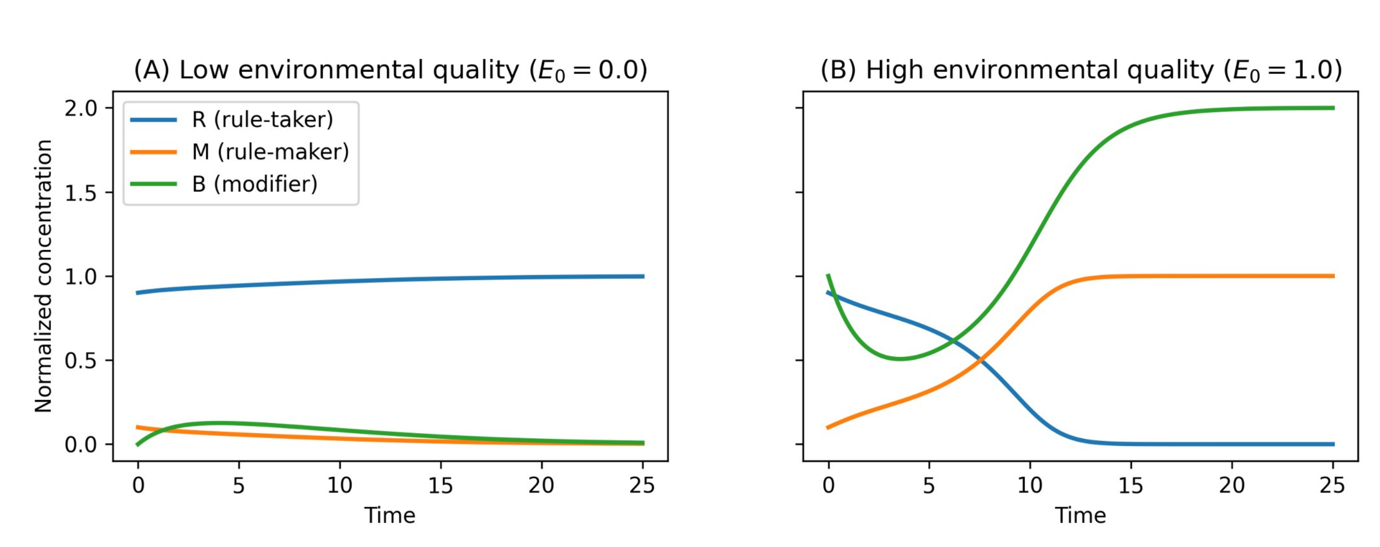
modifier-assisted (catalytic)

$$\mathbf{M} + \mathbf{S} \rightarrow \mathbf{M} + \mathbf{B}$$

modifier production (costly)

$$\mathbf{B} \rightarrow \varnothing$$

modifier decay



TAKE HOME

Three ideas to keep

01

Persistence creates inheritance

Modifiers that outlast a generation transmit conditions, not genes — a chemistry-level form of heredity.

02

History opens basins

A favourable environment lowers the threshold for rule-makers — even when they start rare.

03

Autonomy is conditional

Bistability is not guaranteed: feedback must be strong, persistent, and seeded by the right history.

CLOSING THOUGHT

Replication makes a chemistry persist.
Feedback lets it set the conditions for its own persistence.

That second step is rare — and may be where chemistry began to look like life.

Thank you.

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